

Algorithms Problem Solving

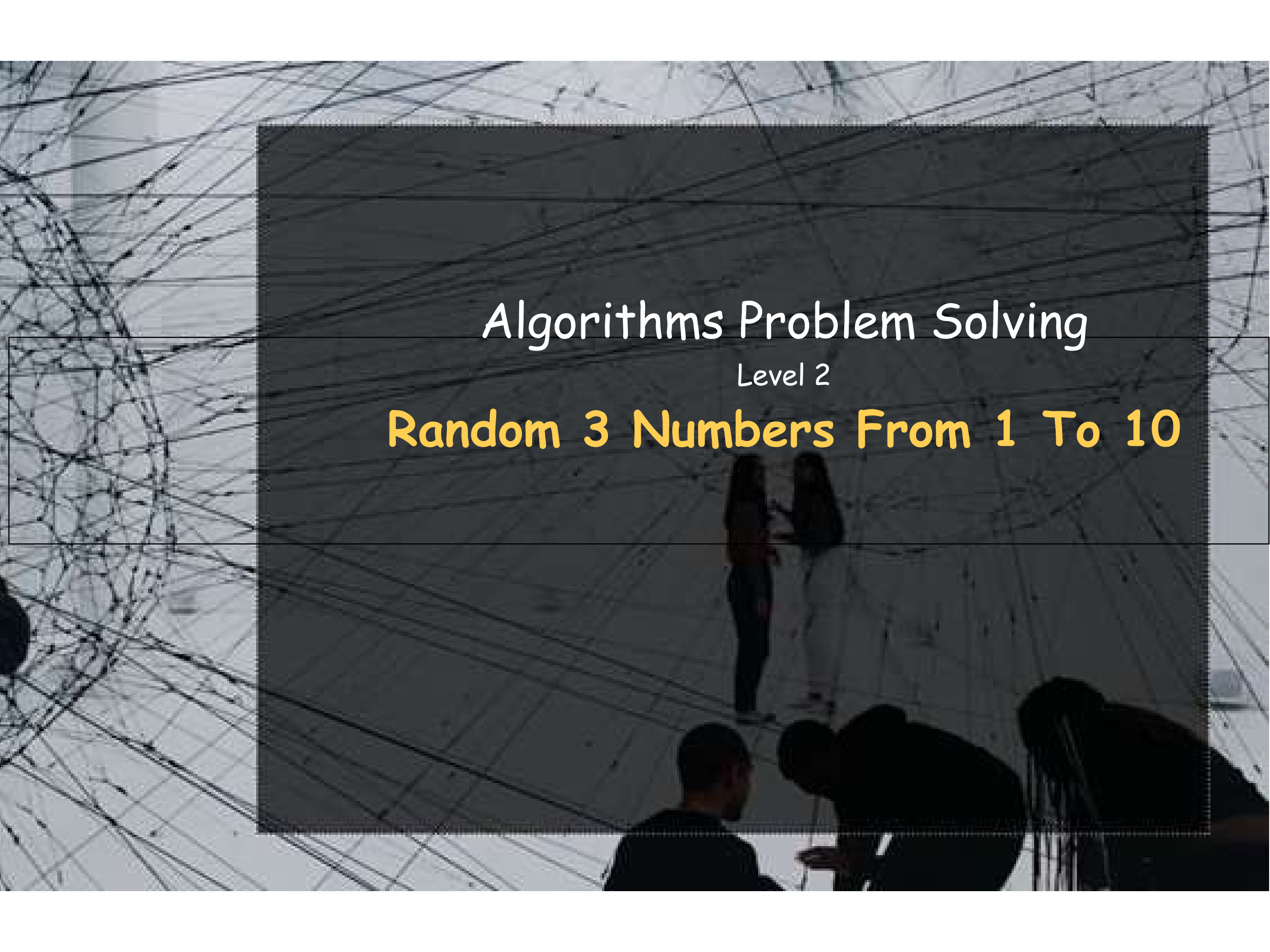
Level 2

Random 3 Numbers From 1
To 10

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Definition

- ◆ A program that generates and prints three random numbers between `1` and `10`.
- ◆ The program uses `rand()` to generate random numbers and `srand()` to initialize the random sequence.

Explanation

- ◆ ``RandomNumber(int from, int to)`` is a function that generates a random number within a specified range.
- ◆ ``rand()`` generates a pseudo-random integer.
- ◆ ``% (to - from + 1)`` limits the result to the required range.
- ◆ ``+ from`` shifts the result so that the minimum value becomes ``from``.
- ◆ In this example:
 - ``from = 1``
 - ``to = 10``
- ◆ Therefore:
 - ``rand() % (10 - 1 + 1) + 1``
 - ``rand() % 10 + 1``
- ◆ The possible results are from ``1`` to ``10``.
- ◆ ``srand((unsigned)time(NULL))`` uses the current time as the seed, helping produce a different sequence each time the program runs.

Steps in main()

1. Initialize the random number generator using `srand()`.
2. Call `RandomNumber(1, 10)`.
3. Generate a random number using `rand()`.
4. Use `%` to limit the result to the required range.
5. Add `from` to shift the result to the correct starting value.
6. Print the generated number.
7. Repeat the function call two more times.

💡 Example

- ◆ The program may produce:

3

5

4

- ◆ Another execution may produce different numbers, such as:

8

1

7

- ◆ The exact output is not fixed because the numbers are generated randomly.

➡ Result

- ◆ Three random numbers between `1` and `10` are printed.
- ◆ Each number can be any value from `1` through `10`.
- ◆ The three numbers do not have to be different.

Note

- ◆ The formula:
 `rand() \% (to - from + 1) + from`
is used to generate a random number within an inclusive range.
- ◆ For ``1`` to ``10``, the formula becomes:
 `rand() \% 10 + 1`
- ◆ ``srand()`` should normally be called once at the beginning of the program, not every time ``RandomNumber()`` is called.
- ◆ The function can be reused with other ranges, for example:
``RandomNumber(5, 20)`` → generates a number from ``5`` to ``20``.

⊘ Warning

- ◆ ``rand()'` generates pseudo-random numbers, not true random numbers.
- ◆ The code uses ``rand()'`, ``srand()'`, and ``time()'` but does not explicitly include ``<stdlib>'` and ``<ctime>'`. Some compilers may still compile it because of indirect includes, but the correct practice is to include the required headers.
- ◆ The three generated numbers can be the same because there is no check to prevent duplicates.

Comparison

- ▣ ``rand()`` → generates the next pseudo-random number.
- ▣ ``srand()`` → sets the starting seed for the random sequence.
- ▣ ``time(NULL)`` → provides the current time, commonly used as a changing seed.
- ▣ ``RandomNumber(1, 10)`` → generates a number from ``1`` to ``10``.

✦ Summary

- ✓ ``rand()`` generates a pseudo-random number.
- ✓ ``srand()`` initializes the random sequence.
- ✓ ``time(NULL)`` provides a changing seed.
- ✓ ``rand() % (to - from + 1) + from`` generates a number between ``from`` and ``to``.
- ✓ ``RandomNumber(1, 10)`` generates numbers from ``1`` to ``10``.
- ✓ The program prints three random numbers.
- ✓ The numbers can be repeated.

Thank you 😊